10.3: The light reactions convert solar energy to the chemical energy of ATP and NADPH

Chloroplasts are chemical factories powered by the sun. Their thylakoids transform light energy into the chemical energy of ATP and NADPH. To better understand the conversion of light to chemical energy, we need to know about some important properties of light.

The Nature of Sunlight

Light is a form of energy known as electromagnetic energy, also called electromagnetic radiation. Electromagnetic energy travels in rhythmic waves analogous to those created by dropping a pebble into a pond. Electromagnetic waves, however, are disturbances of electric and magnetic fields rather than disturbances of a material medium such as water.

The distance between the crests of electromagnetic waves is called the **wavelength**. Wavelengths range from less than a nanometer (for gamma rays) to more than a kilometer (for radio waves). This entire range of radiation is known as the **electromagnetic spectrum(Figure 10.6)**. The segment most important to life is the narrow band from about 380 nm to 740 nm in wavelength. This radiation is known as **visible light** because it can be detected as various colors by the human eye.

Figure 10.6

```
The electromagnetic spectrum.
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The electromagnetic spectrum.

White light is a mixture of all wavelengths of visible light. A prism can sort white light into its component colors by bending light of different wavelengths at different angles. (Droplets of water in the atmosphere can act as prisms, causing a rainbow to form.) Visible light drives photosynthesis.

The electromagnetic wavelength spectrum consists of gamma rays (10 to the minus 5 power nanometers to 10 raised to the power minus 3 power nanometers), X-rays (10 raised to the power minus 3 power nanometers to 1 nanometer), U V light (1 nanometer to 380 nanometers), Visible light (380 nanometers to 750 nanometers), Infrared (750 nanometers to 10 raised to the power 6 nanometers), Microwaves (10 raised to the power 6 nanometers to 1 meter), and Radio Waves (1 meter to 10 raised to the power 3 meters). In the

visible light spectrum, short-wavelength light has higher energy and longer-wavelength light has lower energy.

The model of light as waves explains many of light's properties, but in certain respects light behaves as though it consists of discrete particles, called **photons**. Photons are not tangible objects, but they act like objects in that each of them has a fixed quantity of energy. The amount of energy is inversely related to the wavelength of the light: The shorter the wavelength, the greater the energy of each photon of that light. Thus, a photon of violet light packs nearly twice as much energy as a photon of red light (see Figure 10.6).

Although the sun radiates the full spectrum of electromagnetic energy, the atmosphere acts like a selective window, allowing visible light to pass through while screening out a substantial fraction of other radiation. The part of the spectrum we can see—visible light—is also the radiation that drives photosynthesis.

Photosynthetic Pigments: The Light Receptors

When light meets matter, it may be reflected, transmitted, or absorbed. Substances that absorb visible light are known as *pigments*. Different pigments absorb light of different wavelengths, and the wavelengths that are absorbed disappear. If a pigment is illuminated with white light, the color we see is the color most reflected or transmitted by the pigment. (If a pigment absorbs all wavelengths, it appears black.) We see green when we look at a leaf because chlorophyll absorbs violet-blue and red light while transmitting and reflecting green light **(Figure 10.7)**. The ability of a pigment to absorb various wavelengths of light can be measured with an instrument called a **spectrophotometer**. This machine directs beams of light of different wavelengths through a solution of the pigment and measures the fraction of the light transmitted at each wavelength. A graph plotting a pigment's light absorption versus wavelength is called an **absorption spectrum(Figure 10.8)**.

Figure 10.7

```
    Why leaves are green: interaction of light with
chloroplasts.
```

The chlorophyll molecules of chloroplasts absorb violet blue and red light (the colors most effective in driving photosynthesis) and reflect or transmit green light.

The chlorophyll molecules of chloroplasts absorb violet-blue and red light (the colors most effective in driving photosynthesis) and reflect or transmit green light. This is why leaves appear green.

Animation

Light Energy and Pigments

image

Play

image

Play

**

****Figure 10.8

Research Method: Determining an Absorption Spectrum

Application

An absorption spectrum is a visual representation of how well a particular pigment absorbs different wavelengths of visible light. Absorption spectra of various chloroplast pigments help scientists decipher the role of each pigment in a plant.

Technique

A spectrophotometer measures the relative amounts of light of different wavelengths absorbed and transmitted by a pigment solution.

- ******STEP 1:**White light is separated into colors (wavelengths) by a prism.
- ******STEP 2:**One by one, the different colors of light are passed through the sample (chlorophyll in this example). Green light and blue light are shown here.
- ******STEP 3:**The transmitted light strikes a photoelectric tube, which converts the light energy to electricity.
- ******STEP 4:**The electric current is measured by a galvanometer. The meter indicates the fraction of light transmitted through the sample, from which we can determine the amount of light absorbed.

A diagram of a spectrometer.

The slit on the refracting prism is moved to pass the green light (step 1). After passing through the chlorophyll solution (step 2) and photoelectric tube (step 3), the galvanometer measures high transmittance (low absorption). This reading indicates that chlorophyll absorbs very little green light.

For blue light. The slit on the refracting prism is moved to pass blue light (step 1). After passing through the chlorophyll solution (step 2) and photoelectric tube (step 3), the galvanometer measures low transmittance (high absorption). This reading indicates that chlorophyll absorbs most blue light.

Results

See Figure 10.9a for absorption spectra of three types of chloroplast pigments.

The absorption spectra of chloroplast pigments provide clues to the relative effectiveness of different wavelengths for driving photosynthesis since light can perform work in chloroplasts only if it is absorbed. **Figure 10.9a** shows the absorption spectra of three types of pigments in chloroplasts: **chlorophyll*a***, the key light-capturing pigment that participates directly in the light reactions; the accessory pigment **chlorophyll*b***; and a separate group of accessory pigments called carotenoids. The spectrum of chlorophyll a suggests that violet-blue and red light work best for photosynthesis, since they are absorbed, while green is the least effective color. This is confirmed by an **action spectrum** for photosynthesis **(Figure 10.9b)**, which profiles the relative effectiveness of different wavelengths of radiation in driving the process. An action spectrum is prepared by illuminating chloroplasts with light of different colors and then plotting wavelength against some measure of photosynthetic rate, such as CO_2 consumption or O_2 release. The action spectrum for photosynthesis was first demonstrated by Theodor W. Engelmann, a German botanist, in 1883. Before equipment for measuring O_2 levels had even been invented, Engelmann performed a clever experiment in which he used O_2-requiring bacteria to measure rates of photosynthesis in filamentous algae **(Figure 10.9c)**. His results are a striking match to the modern action spectrum shown in Figure 10.9b.

Figure 10.9

Inquiry: Which wavelengths of light are most effective in driving photosynthesis?

Experiment

Absorption and action spectra, along with a classic experiment by Theodor W. Engelmann, reveal which wavelengths of light are photosynthetically important.

Results

Three graphs of absorption and action spectra.

Part a. Absorption spectra. The three curves are shown in the graph showing the wavelength of light best absorbed by three types of chloroplast pigments. Chlorophyll a absorbs light with wavelengths between 400 to 450 nanometers with a peak at about 440 nanometers. It also absorbs light with wavelengths between 640 and 700 nanometers with a peak at 660 nanometers. Chlorophyll b absorbs light with wavelengths between 400 to 500 nanometers with a peak at about 450 nanometers. It also absorbs light with wavelengths between 620 and 660 nanometers with a peak at 640 nanometers. Carotenoids absorb light with wavelengths between 400 to 520 nanometers with two peaks, one at 450 and one at 475 nanometers.

Three graphs of absorption and action spectra.

Part b. Action spectrum. This graph plots the rate of photosynthesis versus wavelength. The resulting action spectrum resembles the absorption spectrum for chlorophyll a but does not match exactly (see part a). This is partly due to the absorption of light by accessory pigments such as chlorophyll b and carotenoids. The x-axis has wavelength of light (nanometers) ranging from 400 to 700 in increments of 50 units. The y-axis has rate of photosynthesis (measured by oxygen release). The rate of photosynthesis is highest between wavelengths of 400 to 500 nanometers and then drops to a minimum at 560 nanometers. It rises again between 600 to 700 nanometers and peaks at about 680 nanometers.

Three graphs of absorption and action spectra.

Part c. Engelmann's experiment. In 1883, Theodor W. Engelmann shined light through a prism onto an alga, exposing different segments of alga to different wavelength. He used aerobic bacteria, which congregate near an oxygen source, to determine which segments of the algae were releasing the most O 2 and thus photosynthesizing most. Graph of Engelmann's experiment showing the concentration of aerobic bacteria along a filament of algae exposed to different wavelengths of light ranging from 400 to 700 nanometers. Aerobic bacteria congregated in greatest numbers around the parts of the alga illuminated with violet-blue (400 to 500 nanometers) or red light (600 to 700 nanometers).

Conclusion

The action spectra, confirmed by Engelmann's experiment, show which portions of the spectrum are most effective in driving photosynthesis.

Data from T. W. Engelmann, *Bacterium photometricum*. Ein Beitrag zur vergleichenden Physiologie des Licht-und Farbensinnes, *Archiv. für Physiologie* 30:95–124 (1883).

Instructors: A related Experimental Inquiry Tutorial can be assigned in Mastering Biology.

INTERPRET THE DATA *According to the graph, which wavelengths of light drive the highest rates of photosynthesis?*

Notice by comparing Figures 10.9a and 10.9b that the action spectrum for photosynthesis is much broader than the absorption spectrum of chlorophyll a. The absorption spectrum of chlorophyll a alone underestimates the effectiveness of certain wavelengths in driving photosynthesis. This is partly because accessory pigments with different absorption spectra—including chlorophyll b and carotenoids—broaden the spectrum of colors that can be used for photosynthesis. **Figure 10.10** shows the structure of chlorophyll a compared with that of chlorophyll b. A slight structural difference between them is enough to cause the two pigments to absorb at slightly different wavelengths in the red and blue parts of the spectrum (see Figure 10.9a). As a result, chlorophyll a appears blue green and chlorophyll b olive green under visible light.

Figure 10.10

Structure of chlorophyll a and b.

Structure of chlorophyll molecules in chloroplasts of plants.

Structure of chlorophyll molecules in chloroplasts of plants. Chlorophyll a and chlorophyll b differ only in one of the functional groups bonded to the porphyrin ring C H 3 in Chlorophyll a and C H O in chlorophyll b. The porphyrin ring is the light absorbing head of the molecule and includes one magnesium atom at the center. The hydrocarbon tail interacts with hydrophobic regions of proteins inside thylakoid membranes of chloroplasts.

Animation

Space-Filling Model of Chlorophyll

image

Play

image

Play

In the last decade, researchers have identified two other forms of chlorophyll—chlorophyll d and chlorophyll f—that absorb higher wavelengths of light. In 2018, researchers grew a species of cyanobacterium called *Chroococcidiopsis thermalis* under only far-red light with a wavelength of 750 nm.

They concluded that chlorophyll f was functioning in place of chlorophyll a, enabling this cyanobacterium to flourish in very shaded conditions. Higher-wavelength light has less energy, so this observation extends the lower limit of energy (the “red limit”) needed for photosynthesis to occur.

A micrograph showing different photosynthetic pigments in a colony of cyanobacteria.

A colored micrograph of Chroococcidiopsis thermalis showing different types of chlorophyll that absorb different wavelengths of light. Purplish-pink-colored cells possess chlorophyll a and absorb 400 to 700 nanometers of light. Yellow-colored cells possess chlorophyll f, which is acclimated to absorbing higher wavelengths of light (750 nanometers).

Colony of the cyanobacteria*Chroococcidiopsis thermalis*. In this micrograph of cells becoming acclimated to higher wavelengths of light, cells that haven’t yet acclimated and are still using chlorophyll *a* for photosynthesis appear purplish-pink, and cells that have acclimated and are using chlorophyll f appear yellow.

Other accessory pigments include **carotenoids**, hydrocarbons that are various shades of yellow and orange because they absorb violet and blue-green light (see Figure 10.9a). Carotenoids may broaden the spectrum of colors that can drive photosynthesis. However, a more important function of at least some carotenoids seems to be *photoprotection*: These compounds absorb and dissipate excessive light energy that would otherwise damage chlorophyll or interact with oxygen, forming reactive oxidative molecules that are dangerous to the cell. Interestingly, carotenoids similar to the photoprotective ones in chloroplasts have a photoprotective role in the human eye. (Carrots, known for aiding night vision, are rich in carotenoids.) These and related molecules are, of course, found naturally in many vegetables and fruits. They are also often advertised in health food products as “phytochemicals” (from the Greek *phyton*, plant), some of which have antioxidant properties. Plants can synthesize all the antioxidants they require, but humans and other animals must obtain some of them from their diets.

Excitation of Chlorophyll by Light

What exactly happens when chlorophyll and other pigments absorb light? The colors corresponding to the absorbed wavelengths disappear from the spectrum of the transmitted and reflected light, but energy cannot disappear. When a molecule absorbs a photon of light, one of the molecule’s electrons is elevated to an orbital where it has more potential energy (see Figure 2.6b). When the electron is in its normal orbital, the pigment molecule is said to be in its ground state. Absorption of a photon boosts an electron to an orbital of higher energy, and the pigment molecule is then said to be in an excited state. The only photons absorbed are those whose energy is exactly equal to the energy difference between the ground state and an excited state, and this energy difference varies from one kind of molecule to another. Thus, a particular compound

absorbs only photons corresponding to specific wavelengths, which is why each pigment has a unique absorption spectrum.

Once absorption of a photon raises an electron to an excited state, the electron cannot stay there long **(Figure 10.11a)**. The excited state, like all high-energy states, is unstable. Generally, when isolated pigment molecules absorb light, their excited electrons drop back down to the ground-state orbital in a billionth of a second, releasing their excess energy as heat. This conversion of light energy to heat is what makes the top of an automobile so hot on a sunny day. (White cars are coolest because their paint reflects all wavelengths of visible light.) In isolation, some pigments, including chlorophyll, emit light as well as heat after absorbing photons. As excited electrons fall back to the ground state, photons are given off, an afterglow called fluorescence. An illuminated solution of chlorophyll isolated from chloroplasts will fluoresce in the red part of the spectrum and also give off heat **(Figure 10.11b)**. This is best seen by illuminating with ultraviolet light, which chlorophyll can also absorb (see Figures 10.6 and 10.9a). Viewed under visible light, the fluorescence would be difficult to see against the green of the solution.

Figure 10.11

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Excitation of isolated chlorophyll by light.
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Excitation of isolated chlorophyll molecule.

(a) *Absorption of a photon causes a transition of the chlorophyll molecule from its ground state to its excited state. The photon boosts an electron to an orbital where it has more potential energy. If the illuminated molecule exists in isolation, its excited electron immediately drops back down to the ground-state orbital, and its excess energy is given off as heat and fluorescence (light).* ***(b)*** *A chlorophyll solution excited with ultraviolet light fluoresces with a red-orange glow.*

In Part a, a photo hits a chlorophyll molecule. An electron goes from ground state to an excited state with an increase in energy. As it falls back to the ground state, it gives off heat and a photon as fluorescence.

Part b is a beaker of fluorescence, or a liquid with a red-orange glow.

WHAT IF?***If a leaf containing the same concentration of chlorophyll as in the solution was exposed to the same ultraviolet light, no fluorescence would be seen. Propose an explanation for the difference in fluorescence emission between the solution and the leaf.*

*For suggested answer, see*Appendix A.**

A Photosystem: A Reaction-Center Complex Associated with Light-Harvesting Complexes

Chlorophyll molecules excited by the absorption of light energy produce very different results in an intact chloroplast than they do in isolation (see Figure 10.11). In their native environment of the thylakoid membrane, chlorophyll molecules are organized along with other small organic molecules and proteins into complexes called photosystems.

A **photosystem** is composed of a reaction-center complex surrounded by several light-harvesting complexes **(Figure 10.12)**. The **reaction-center complex** is an organized association of proteins holding a special pair of chlorophyll a molecules and a primary electron acceptor. Each **light-harvesting complex** consists of various pigment molecules (which may include chlorophyll a, chlorophyll b, and multiple carotenoids) bound to proteins. The number and variety of pigment molecules enable a photosystem to harvest light over a larger surface area and a larger portion of the spectrum than could any single pigment molecule alone. Together, these light-harvesting complexes act as an antenna for the reaction-center complex. When a pigment molecule absorbs a photon, the energy is transferred from pigment molecule to pigment molecule within a light-harvesting complex, like a human “wave” at a sports arena, until it is passed to the pair of chlorophyll a molecules in the reaction-center complex. This pair of chlorophyll a molecules is special because their molecular environment—their location and the other molecules with which they are associated—enables them to use the energy from light not only to boost one of their electrons to a higher energy level, but also to transfer it to a different molecule—the **primary electron acceptor**, which is a molecule capable of accepting electrons and becoming reduced.

Figure 10.12

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The structure and function of a photosystem.
```

The structure and function of a photosystem.

Part a. How a photosystem harvests light. When a photon strikes a pigment molecule in a light-harvesting complex, the energy is passed from molecule to molecule until it reaches the reaction-center complex. Here, an excited electron from the special pair of chlorophyll molecules is transferred to the primary electron acceptor.

Part b. Structure of a photosystem. This computer model, based on X-ray crystallography, shows two photosystem complexes side by side. Chlorophyll molecules (small green ball-and-stick models) are

interspersed with protein subunits (cylinders and ribbons). For simplicity, this photosystem will be shown as a single complex in the rest of the chapter.

Figure B is a diagram of the structure of a photosystem.

A computer model, based on X-ray crystallography, shows two photosystem complexes side by side. Chlorophyll and protein subunits are interspersed in a thylakoid membrane between stroma and thylakoid space. Many alpha helices span the membrane. For simplicity, a photosystem will be shown as a single complex in the rest of the chapter.

The solar-powered transfer of an electron from the reaction-center chlorophyll a pair to the primary electron acceptor is the first step of the light reactions. As soon as the chlorophyll electron is excited to a higher energy level, the primary electron acceptor captures it; this is a redox reaction. In the flask shown in Figure 10.11b, isolated chlorophyll fluoresces because there is no electron acceptor, so electrons of photoexcited chlorophyll drop right back to the ground state. In the structured environment of a chloroplast, however, an electron acceptor is readily available, and the potential energy represented by the excited electron is not dissipated as light and heat. Thus, each photosystem—a reaction-center complex surrounded by light-harvesting complexes—functions in the chloroplast as a unit. It converts light energy to chemical energy, which will ultimately be used for the synthesis of sugar.

The thylakoid membrane is populated by two types of photosystems that cooperate in the light reactions of photosynthesis: **photosystem II(PS II)** and **photosystem I(PS I)**. (They were named in order of their discovery, but photosystem II functions first in the light reactions.) Each has a characteristic reaction-center complex—a particular kind of primary electron acceptor next to a special pair of chlorophyll a molecules associated with specific proteins. The reaction-center chlorophyll a of photosystem II is known as P680 because this pigment is best at absorbing light having a wavelength of 680 nm (in the red part of the spectrum). The chlorophyll a at the reaction-center complex of photosystem I is called P700 because it most effectively absorbs light of wavelength 700 nm (in the far-red part of the spectrum). These two pigments, P680 and P700, are nearly identical chlorophyll a molecules. However, their association with different proteins in the thylakoid membrane affects the electron distribution in the two pigments and accounts for the slight differences in their light-absorbing properties. Now let's see how the two types of photosystems work together in using light energy to generate ATP and NADPH, the two main products of the light reactions.

Linear Electron Flow

Light drives the synthesis of ATP and NADPH by energizing the two types of photosystems embedded in the thylakoid membranes of chloroplasts. The key to this energy transformation is a flow of electrons through

the photosystems and other molecular components built into the thylakoid membrane. This is called **linear electron flow**, and it occurs during the light reactions of photosynthesis, as shown in **Figure 10.13**. The numbered steps in the text correspond to the numbered steps in the figure.

Figure 10.13

```
    How linear electron flow during the light reactions
generates ATP and NADPH.
```

How linear electron flow during the light reactions generates A T P and N A D P H.

A photon of light strikes one of the pigment molecules in a light-harvesting complex of P S 2, boosting one of its electrons of a higher energy level. A primary electron acceptor relays the electron through the electron transport chain, which gives off A T P. This chain leads to P S 1, hitting P 700. More light hits pigment molecules in this system, which also leads to P 700. This excites an electron to a primary electron acceptor, which then relays it into another part of the electron transport chain where N A D P plus reductase releases N A D P H.

The gold arrows trace the flow of light-driven electrons from water to NADPH. The black arrows trace the transfer of energy from pigment molecule to pigment molecule. To see these proteins in their cellular context, see Figure 6.32b.

Figure 10.13 Walkthrough

How linear electron flow during the light reactions generates ATP and NADPH.

Watch an explanation of this figure in the Figure 10.13 Walkthrough: How linear electron flow during the light reactions generates ATP and NADPH.

image

Play

image

**

- ******STEP 1:**A photon of light strikes one of the pigment molecules in a light-harvesting complex of PS II, boosting one of its electrons to a higher energy level. As this electron falls back to its ground state, an electron in a nearby pigment molecule is simultaneously raised to an excited state. The process continues, with the energy being

relayed to other pigment molecules until it reaches the P680 pair of chlorophyll a molecules in the PS II reaction-center complex. It excites an electron in this pair of chlorophylls to a higher energy state.

- ******STEP 2:**This electron is transferred from the excited P680 to the primary electron acceptor. We can refer to the resulting form of P680, missing the negative charge of an electron, as P680+.

- ******STEP 3:**An enzyme catalyzes the splitting of a water molecule into two electrons, two hydrogen ions (H+), and an oxygen atom. The electrons are supplied one by one to the P680+ pair, each electron replacing one transferred to the primary electron acceptor. (P680+ is the strongest biological oxidizing agent known; its electron "hole" must be filled. This greatly facilitates the transfer of electrons from the split water molecule.) The H+ are released into the thylakoid space (interior of the thylakoid). The oxygen atom immediately combines with an oxygen atom generated by the splitting of another water molecule, forming O2.

- ******STEP 4:**Each photoexcited electron passes from the primary electron acceptor of PS II to PS I via an electron transport chain, the components of which are similar to those of the electron transport chain that functions in cellular respiration. The electron transport chain between PS II and PS I is made up of the electron carrier plastoquinone (Pq), a cytochrome complex, and a protein called plastocyanin (Pc). Each component carries out redox reactions as electrons flow down the electron transport chain, releasing free energy that is used to pump protons (H+) into the thylakoid space, contributing to a proton gradient across the thylakoid membrane.

- ******STEP 5:**The potential energy stored in the proton gradient is used to make ATP in a process called chemiosmosis, to be discussed shortly.

- ******STEP 6:**Meanwhile, light energy has been transferred via light-harvesting complex pigments to the PS I reaction-center complex, exciting an electron of the P700 pair of chlorophyll a molecules located there. The photoexcited electron is then transferred to PS I's primary electron acceptor, creating an electron "hole" in the P700—which we now can call P700+. In other words, P700+ can now act as an electron acceptor, accepting an electron that reaches the bottom of the electron transport chain from PS II.

- ******STEP 7:**Photoexcited electrons are passed in a series of redox reactions from the primary electron acceptor of PS I down a second electron transport chain through the protein ferredoxin (Fd). (This chain does not create a proton gradient and thus does not produce ATP.)

- ******STEP 8:**The enzyme NADP+ reductase catalyzes the transfer of electrons from Fd to NADP+. Two electrons are required for its reduction to NADPH. Electrons in NADPH are at a higher energy level than they are in water (where they started), so they are more readily available for the reactions of the Calvin cycle. This process also removes an H+ from the stroma.

The energy changes of electrons during their linear flow through the light reactions are shown in a mechanical analogy in **Figure 10.14**. Although the scheme shown in Figures 10.13 and 10.14 may seem complicated, do not lose track of the big picture: The light reactions use solar power to generate ATP and NADPH, which provide chemical energy and reducing power, respectively, to the carbohydrate-synthesizing reactions of the Calvin cycle.

Figure 10.14

```
    A mechanical analogy for linear electron flow during the
light reactions.
```

A mechanical analogy for linear electron flow during the light reactions.

A worker uses a photon hammer to strike a lever which sends the electron from the ground level to the top of the Photosystem II tower which is 3 platforms high. Another worker catches the electron at the top of the platform and then puts it on the mill conveyor flowing down to the first platform of the Photosystem I tower. As the electron travels down the conveyor, the mill makes A T P, and then A T P is released. The electron continues down the conveyor to the first platform of the Photosystem I tower. Another worker uses a photon hammer to strike a lever that sends the electron back up to the top of the fourth platform of the Photosystem I tower where a worker catches the electron and deposits it in the N A D P H bucket.

Cyclic Electron Flow

In certain cases, photoexcited electrons can take an alternative path called **cyclic electron flow**, which uses photosystem I but not photosystem II. You can see in **Figure 10.15** that cyclic flow is a short circuit:

The electrons cycle back from ferredoxin (Fd) to the cytochrome complex, then via a plastocyanin molecule (Pc) to a P700 chlorophyll in the PS I reaction-center complex. There is no production of NADPH and no release of oxygen that results from this process. On the other hand, cyclic flow does generate ATP.

Figure 10.15

Cyclic electron flow.

Cyclic electron flow.

Photoexcited electrons from Photosystem I are occasionally shunted back from ferredoxin (F d) to chlorophyll via the cytochrome complex and plastocyanin (P c). This electron shunt supplements the supply of A T P (via chemiosmosis) but produces no N A D P H. Compared to linear electron flow, cyclic electron flow does not include Photosystem II nor does it include electron transport and N A D P plus reductase.

Photoexcited electrons from PS I are occasionally shunted back from ferredoxin (Fd) to chlorophyll via the cytochrome complex and plastocyanin (Pc). This electron shunt supplements the supply of ATP (via chemiosmosis) but produces no NADPH. The “shadow” of linear electron flow is included in the diagram for comparison with the cyclic route. The two Fd molecules in this diagram are actually one and the same—the final electron carrier in the electron transport chain of PS I—although it is depicted twice to clearly show its role in two parts of the process.

VISUAL SKILLS*Look at**Figure 10.14**and explain how you would alter it to show a mechanical analogy for cyclic electron flow.*

*For suggested answer, see*Appendix A.**

Rather than having both PS II and PS I, several of the currently existing groups of photosynthetic bacteria are known to have a single photosystem related to either PS II or PS I. For these species, which include the purple sulfur bacteria (see Figure 10.2e) and the green sulfur bacteria, cyclic electron flow is the one and only means of generating ATP during the process of photosynthesis. Evolutionary biologists hypothesize that these bacterial groups are descendants of ancestral bacteria in which photosynthesis first evolved, in a form similar to cyclic electron flow.

Cyclic electron flow can also occur in photosynthetic species that possess both photosystems; this includes some prokaryotes, such as the cyanobacteria shown in Figure 10.2d, as well as the eukaryotic photosynthetic species that have been tested thus far. Although the process is probably in part an “evolutionary leftover,” research suggests it plays at least one beneficial role for these organisms. Plants with mutations that render them unable to carry out cyclic electron flow are capable of growing well in low light, but do not grow well where light is intense. This is evidence for the idea that cyclic electron flow may be photoprotective. Later

you'll learn more about cyclic electron flow as it relates to a particular adaptation of photosynthesis (C4 plants; see Concept 10.5).

Whether ATP synthesis is driven by linear or cyclic electron flow, the actual mechanism is the same. Before we move on to consider the Calvin cycle, let's review chemiosmosis, the process that uses membranes to couple redox reactions to ATP production.

A Comparison of Chemiosmosis in Chloroplasts and Mitochondria

Chloroplasts and mitochondria generate ATP by the same basic mechanism: chemiosmosis (see Figure 9.14). An electron transport chain pumps protons (H+) across a membrane as electrons are passed through a series of carriers that have progressively more affinity for electrons. Thus, electron transport chains transform redox energy to a proton-motive force, potential energy stored in the form of an H+ gradient across a membrane. An ATP synthase complex in the same membrane couples the diffusion of hydrogen ions down their gradient to the phosphorylation of ADP, forming ATP.

Some of the electron carriers, including the iron-containing proteins called cytochromes, are very similar in mitochondria and chloroplasts (see Figure 6.32b and c). The ATP synthase complexes of the two organelles are also quite similar. But there are noteworthy differences between photophosphorylation in chloroplasts and oxidative phosphorylation in mitochondria. Both work by way of chemiosmosis, but in chloroplasts, the high-energy electrons dropped down the transport chain come from water, while in mitochondria, they are extracted from organic molecules (which are thus oxidized). Chloroplasts do not need molecules from food to make ATP; their photosystems capture light energy and use it to drive the electrons from water to the top of the transport chain. In other words, mitochondria use chemiosmosis to transfer chemical energy from food molecules to ATP, whereas chloroplasts use it to transform light energy into chemical energy in ATP.

Although the spatial organization of chemiosmosis differs slightly between chloroplasts and mitochondria, it is easy to see similarities in the two **(Figure 10.16)**. Electron transport chain proteins in the inner membrane of the mitochondrion pump protons from the mitochondrial matrix out to the intermembrane space, which then serves as a reservoir of hydrogen ions. Similarly, electron transport chain proteins in the thylakoid membrane of the chloroplast pump protons from the stroma into the thylakoid space, which functions as the H+ reservoir. If you imagine the cristae of mitochondria pinching off from the inner membrane, this may help you see how the thylakoid space and the intermembrane space are comparable spaces in the two organelles, while the mitochondrial matrix is analogous to the stroma of the chloroplast.

Figure 10.16

Comparison of chemiosmosis in mitochondria and chloroplasts.

Comparison of chemiosmosis in mitochondria and chloroplasts.

Comparison of chemiosmosis in mitochondria and chloroplasts. In both kinds of organelles, electron transport chains pump protons (H plus) across a membrane from a region of low H plus concentration (mitochondrial matrix or stroma in the chloroplast) to one of high H plus concentration (intermembrane space in the mitochondrion or thylakoid space in the chloroplast). The protons then diffuse back across the membrane (inner membrane in the mitochondrion or thylakoid membrane in the chloroplast) through A T P synthase, driving the synthesis of A T P.

In both kinds of organelles, electron transport chains pump protons (H+) across a membrane from a region of low H+ concentration (light gray in this diagram) to one of high H+ concentration (dark gray). The protons then diffuse back across the membrane through ATP synthase, driving the synthesis of ATP.

MAKE CONNECTION*Describe how you would change the*pH*in order to artificially cause ATP synthesis (a) outside an isolated mitochondrion (assume*H+*can freely cross the outer membrane; see*Figure 9.14*)* and (b) in the stroma of a chloroplast. Explain.*

*For suggested answer, see*Appendix A*.*

In the mitochondrion, protons diffuse down their concentration gradient from the intermembrane space through ATP synthase to the matrix, driving ATP synthesis. In the chloroplast, ATP is synthesized as the hydrogen ions diffuse from the thylakoid space back to the stroma through ATP synthase complexes **(Figure 10.17)**, whose catalytic knobs are on the stroma side of the membrane. Thus, ATP forms in the stroma, where it is used to help drive sugar synthesis during the Calvin cycle.

Figure 10.17

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The light reactions: organization of the thylakoid membrane.
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The light reactions and chemiosmosis: the organization of the thylakoid membrane.

One. Energy from light excites an electron from chlorophyll, which is transferred to the primary electron acceptor in photosystem 2.

Two. In the thylakoid space, water is split into 2 electrons, which replace chlorophyll's electron twice, and 2 H pluses, which contribute to the high H plus concentration in the thylakoid space, an O atom, which joins another O, forming O 2.

Three. Arrows track the flow of electrons. As electrons travel down the electron transport chain, 4 H plus ions are pumped into the thylakoid space, increasing the concentration of H plus and decreasing the p H.

Four. In chemiosmosis, the diffusion of H plus from the thylakoid space back to the stroma down the H plus gradient powers A T P synthase, producing A T P.

Five. Low-energy electrons from water end up in N A D P H (high-energy electrons). N A D P H formation removes H plus from the stroma, decreasing its H plus concentration, and increasing the p H.

VISUAL SKILLS*Which three steps in the figure contribute to the [H+] gradient across the thylakoid membrane?*

*For suggested answer, see*Appendix A.**

Animation

The Light Reactions

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The proton (H+) gradient, or pH gradient, across the thylakoid membrane is substantial. When chloroplasts in an experimental setting are illuminated, the pH in the thylakoid space drops to about 5 (the H+ concentration increases), and the pH in the stroma increases to about 8 (the H+ concentration decreases). This gradient of three pH units corresponds to a thousandfold difference in H+ concentration. If the lights are then turned off, the pH gradient is abolished, but it can quickly be restored by turning the lights back on. Experiments such as this provided strong evidence in support of the chemiosmotic model.

The currently accepted model for the organization of the light-reaction “machinery” within the thylakoid membrane is based on several research studies. Each of the molecules and molecular complexes in Figure 10.17 is present in numerous copies in each thylakoid. Notice that NADPH, like ATP, is produced on the side of the membrane facing the stroma, where the Calvin cycle reactions take place.

Let’s step back and see the big picture by summarizing the light reactions. Electron flow pushes electrons from water, where they are at a state of low potential energy, ultimately to NADPH, where they are stored at a state of high potential energy. The light-driven electron flow also generates ATP. Thus, the equipment of the thylakoid membrane converts light energy to chemical energy stored in ATP and NADPH. O_2 is produced as a by-product. Let’s now see how the enzymes of the Calvin cycle use the products of the light reactions to synthesize sugar from CO_2.

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